

July 3, 1945.

W. O. BENNETT, JR., ET AL

2,379,780

HAIRSPRING MOUNTING

Filed Aug. 4, 1943

FIG. 1

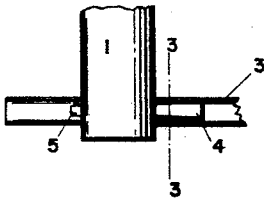


FIG. 2

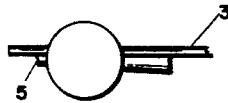


FIG. 3

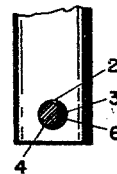


FIG. 4

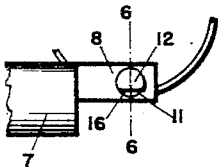


FIG. 5

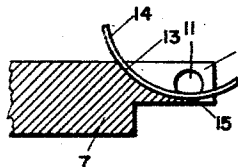


FIG. 6

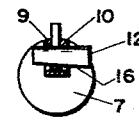


FIG. 7

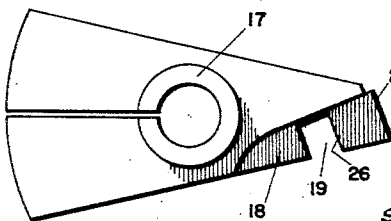


FIG. 8

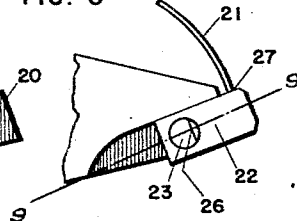


FIG. 9

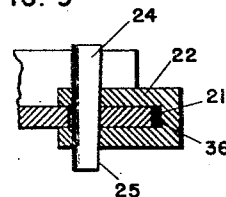


FIG. 10

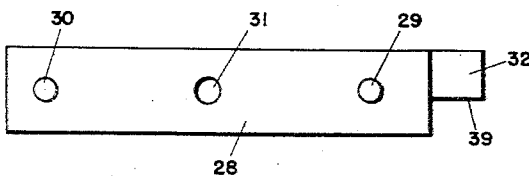


FIG. 11

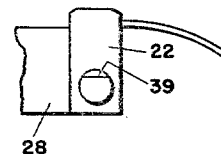


FIG. 12

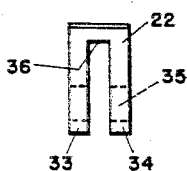


FIG. 13



FIG. 14

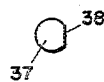
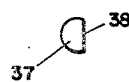


FIG. 15



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2,379,780

HAIRSPRING MOUNTING

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Application August 4, 1943, Serial No. 497,326

5 Claims. (Cl. 58—115)

This invention is directed to a method and apparatus for securing the ends of a timepiece hairspring.

The means formerly used to secure the ends of hairsprings has been universal for all types of timekeeping instruments. This method, however satisfactory in watches, has not proved satisfactory when applied to chronometers or more accurate timepieces. A watch is made with a regulator having two regulator pins which depend on each side of the hairspring and serve as contact points for the hairspring in its oscillations. In the more accurate timepieces or chronometers there is no regulator. The hairspring oscillates from the points at which it is secured to the stud and collet. In the past this anchoring or securing has been accomplished by driving a tapered pin through a round hole in the stud or collet through which the hairspring passes and thus wedging the hairspring by means of the pin.

This old type of anchoring caused the hairspring to curve about a longitudinal axis in accordance with the curvature of the opening and the curvature of the tapered pin. This curvature caused the hairspring to flex more in one direction than another and while not of great importance in a watch because of the regulator it is of extreme importance in the more accurate timepieces. It is the purpose of this invention to set forth a method of anchoring or securing the hairspring so that the flexure is the same in each direction.

It is the object of the present invention to provide apparatus for clamping the hairspring of an accurate timepiece so that the clamping action is in a single plane.

It is a further object of the present invention to provide an apparatus which clamps the hairspring so that the clamping contact is in a plane normal to the length of the hairspring blade.

It is a further object of the present invention to provide means by which the hairspring of an accurate timepiece is anchored or secured so that the point from which it flexes is in a single plane normal to the length of the hairspring blade.

It is a further object of the present invention to provide an apparatus for securing or clamping the ends of a hairspring for an accurate timepiece in such a manner that the clamping action is accomplished without distorting the hairspring from its natural curve.

It is a further object to provide means for clamping the hairspring in the manner of a vise whereby the clamping action is carried out across the entire width of the hairspring.

It is a further object to provide a flat surfaced tapered pin to clamp the hairspring against an opposing flat surface whereby the clamping action is distributed across the entire width of the hairspring and in a single plane normal to the length of the hairspring.

It is a further object to provide a collet or stud designed to receive a hairspring and a tapered pin at right angles to said hairspring to clamp said hairspring in said collet at a single plane normal to the length of said hairspring.

It is a further object to provide a collet or stud to receive a hairspring and a clamping member cooperating with said collet or stud and engaging said hairspring to securely anchor said hairspring.

It is a further object to provide a collet or stud designed to receive a hairspring, with a clamping member cooperating with said collet or stud and a tapered pin acting on said clamping member and said collet or stud to securely anchor said hairspring.

A further object of the present invention is to provide a U-shaped clamping member which engages a portion of the end of a hairspring, the pressure being equally distributed throughout the length and width of said engagement. The invention is illustrated in the accompanying drawing in which:

Fig. 1 is a front elevation of the old method of securing the hairspring.

Fig. 2 is a bottom plan of the old method of securing the hairspring.

Fig. 3 is a sectional line 3—3 of Fig. 1.

Fig. 4 is a front elevation of the stud holding the end of said hairspring.

Fig. 5 is a longitudinal section of Fig. 4 showing the hairspring in dotted lines.

Fig. 6 is a section on line 6—6 of Fig. 4.

Fig. 7 is a top plan of a hairspring collet.

Fig. 8 is a partial view of the collet showing the hairspring in clamped position.

Fig. 9 is longitudinal section of the collet with the hairspring in clamped position.

Fig. 10 is a view of the stud which is attached to the balance wheel.

Fig. 11 is a partial view of the stud shown in Fig. 10 showing the hairspring in clamped position.

Fig. 12 is a view of the U-shaped clamping element.

Fig. 13 is a side view of the tapered pin.

Fig. 14 is a top plan of said pin.

Fig. 15 is a bottom plan of said pin.

The drawing illustrates in Figs. 1 to 3 the old

method of clamping a hairspring which will be described to point out the disadvantages of the old method and the relative advances which applicant has made over said old method. A hairspring stud 1 is formed with an opening 2 through which a hairspring 3 is passed and secured in position by a tapered pin 4, which after being driven in is broken off at 5. This arrangement of securing the hairspring results in a curvature of the hairspring in accordance with the curvature of the opening 2 and the tapered pin 4 and can be seen at 6 in Fig. 3. This curvature causes the hairspring to flex more in one direction than another and results in an uneven period which precludes accurate time. The tapered pin 5 bears against the length of the hairspring for an indeterminate distance, depending how far the pin is driven in the hole, the length of the pin, and the position at which it is broken off. All of these factors enter into the determination of the point at which the hairspring flexes, and all bear on the accuracy of the time-keeping instrument. As has been said, this is not extremely important in watches where a regulator is used which has regulator pins which bear on each side of the hairspring and thus limit its flexure. However, in chronometers there is no regulator and the point at which the hairspring flexes must be absolutely fixed and once secured must remain anchored at the exact same point, which is almost impossible with this old style of fastening.

To be specific, a change in length of $\frac{1}{100,000}$ (five-one hundred thousandth of an inch) in the usual chronometer hairspring will cause a change of rate of $\frac{1}{10}$ (one-tenth) second per day which will demonstrate the extreme importance of absolute fixity of the hairspring.

Figs. 4, 5 and 6 are directed to a type of anchoring where the point of contact is determined by the contact of a wedge pin with that portion of the hairspring stud which supports the hairspring. The hairspring stud 7 is formed with an extending yoke 8 having arms 9 and 10, said arms having a hole 11 to receive a tapered pin 12. The stud between the arms is formed with a curved surface 13 to receive the hairspring 14. This curved portion terminates abruptly at 15 which point is directly under the flattened edge 16 of the tapered pin, thus presenting a securing contact point which is in a single plane normal to the length of the hairspring which insures equal flexure of the hairspring in each direction. The tapered side 16 is exaggerated in the drawing, there being only a difference of five thousandths of an inch in the entire length of the pin. This flat taper gives contact entirely across the width of the hairspring and equal to the width of the pin, thus presenting a securing means which holds the hairspring rigidly at a point which bears across the entire width of the spring and in a single plane normal to the length of the hairspring.

Figs. 7, 8 and 9 are directed to a hairspring collet which is secured to the balance staff and which in turn secures one end of the hairspring. This collet is formed with a hub 17 adapted to engage the balance staff and secure said collet in desired position. A recessed section 18 is formed with a cut-out portion 19 and an end portion 20, against which a hairspring 21 is clamped by a U-shaped clamping member 22. The U-shaped clamping member is formed with an opening 23 passing through its legs and designed when applied to the collet to be directly over the

cut-out portion 19. A wedge pin 24 having a flat tapered side 25 is passed through hole 23, said flat portion 25 engaging the edge 26 of the cut-out portion 19 and pulling the U-shaped clamping member against the hairspring 21 to securely hold or anchor said hairspring in position. Thus the hairspring is securely held throughout a length equal to the width of the U-shaped clamping member 22 and presents a contact point 27 about which the hairspring flexes in its oscillations. This contact point 27 is in a single plane normal to the length of the hairspring. The U-shaped clamping member may be formed either flat as shown in Figs. 8 and 9 or curved to conform to the exact curve of the hairspring.

Figs. 10 and 11 are directed to another form of hairspring stud, one which is adapted to secure the end of the hairspring which is not held by the collet shown in Fig. 7 and to be clamped to some rigid portion of the timepiece. This stud 28 is formed with pins 29 and 30 and a central opening 31 for securing to the balance cock and has a recessed portion 32 to receive the U-shaped clamping member 22. The clamping member is secured to the collet 28 in a manner similar to the way in which it is secured to the collet shown in Figs. 7, 8 and 9.

Fig. 12 shows a U-shaped clamping member 22 having legs 33 and 34 and opening 35 and a hairspring clamping edge 36.

The pin used in conjunction with said U-shaped clamping member is a cylindrical pin 37 having a tapered flattened portion 38 to engage the edge 26 of one collet and 39 of the other collet and so pull the U-shaped clamping member against the hairspring so as to hold it in position.

In use this means of anchoring the hairspring especially in chronometers and accurate timepieces where no regulator is used has enabled the hairspring to be assembled without the distortion which heretofore had to be overcome by trial and error to provide an accurate running chronometer. With the apparatus as set forth in this application the hairspring is set up without distortion and generally requires no adjustment after assembly.

What is claimed is:

1. A hairspring clamp comprising a collet formed with a recessed portion, a shoulder for said portion, a U-shaped clamping member engaging said recessed portion and receiving said hairspring, means for clamping said hairspring between said U-shaped member and said shoulder, where said clamping point is in a single plane, normal to the length of said hairspring.

2. A hairspring clamp comprising a collet formed with a recessed portion, said recessed portion having a rectangular shaped opening therein, a U-shaped clamping member engaging said recessed portion and receiving the hairspring between the end of the recessed portion and the U-shaped part of said clamping member, a wedge pin entering a hole formed in said U-shaped clamping member and engaging the edge of said rectangular shaped opening of the recessed portion to securely clamp the said hairspring.

3. An apparatus of the class described comprising a collet formed with a recessed section having a flat end portion, a U-shaped clamping member fitting over and cooperating with the flat end of the recessed section to receive the end of a hairspring therebetween and means for

securing the U-shaped member in clamping position.

4. An apparatus of the class described comprising a collet formed with a recessed section having a flat end portion, a U-shaped clamping member cooperating with the flat end of said recessed section to receive the end of a hairspring therebetween, said U-shaped member being formed with a hole, and a tapered pin adapted to pass through the hole in the U-shaped member and engage a portion of the recessed section to secure the U-shaped member in clamping position.

5. A hairspring clamp comprising a stud formed with a groove having opposed sides formed with a hole at right angles to the groove, the bottom surface of said groove being curved in conformity with the curve of a hairspring, a tapered pin receivable in said hole for securing a hairspring at the termination of the groove, said securing being in a single plane normal to the length of the hairspring to permit unrestricted equal flexure of the hairspring in either direction from said securing point.

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